

Calculations

$$P = 400 \text{ lbf}$$

$$\delta = 0.009 \text{ in}$$

$$E = 10 \times 10^6 \text{ psi}$$

$$S_y = 40 \text{ ksi}$$

$$w = 1.0 \text{ in}$$

$$h = 1.0 \text{ in}$$

$$t = 0.1 \text{ in}$$

$$A = w \cdot h - (w - 2t)(h - 2t) = 1.0(1.0) - (0.8)(0.8) = 0.36 \text{ in}^2$$

$$L = \frac{\delta \cdot A \cdot E}{P} = \frac{0.009 \cdot 0.36 \cdot 10,000,000}{400} = 81 \text{ in}$$

Analysis

$$\sigma_{\max} = 1.35 \text{ ksi (from the simulation)}$$

$$SF = \frac{40 \text{ ksi}}{1.35 \text{ ksi}} = 29.62962963$$

$$\text{FEA Result} = 0.008997 \text{ in}$$

$$\% \text{ diff} = \frac{|0.009 - 0.008997|}{\frac{0.009 + 0.008997}{2}} \cdot 100 = 0.033\% \text{ difference}$$

$$K_t = 2.5 \quad 0.2 \text{ pin} \quad 1 \text{ in flat face} \quad d/w = 0.2 \quad 0.32 \text{ in}^2$$

$$\sigma_{\text{net}} = \frac{400}{0.32} = 1250 \text{ psi} = 1.25 \text{ ksi}$$

$$\sigma_{\text{peak}} = K_t \cdot \sigma_{\text{net}} = 2.5 \cdot 1.25 = 3.125 \text{ ksi}$$

$$SF = \frac{40}{3.125} = 12.8 \checkmark$$